

Health Insurance Charges Predictors Using Machine Learning

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GitHub Repo: <https://github.com/romeor26/STATS-507>

Abstract—This study explores the use of machine learning to predict health insurance charges based on demographic and lifestyle data. By leveraging regression models and Random Forest, we identify key predictors such as smoking status, BMI, and age. Statistical p-value testing further highlights the significance of predictors, while permutation importance demonstrates predictor contributions in a non-linear framework. This work showcases the complementary role of statistical and machine learning methods in improving pricing models and generating actionable insights for health insurance.

I. INTRODUCTION

A. Background and Motivation

Healthcare costs have risen significantly, burdening individuals and families worldwide. Accurate pricing in health insurance is crucial for managing risks and ensuring fairness. Traditionally, premiums are calculated using demographic and lifestyle features such as age, BMI, smoking status, and region [1]. However, traditional methods often fail to capture complex interactions among these variables.

Recent advancements in machine learning offer opportunities to improve prediction accuracy while uncovering hidden patterns in data [2]. This project leverages statistical testing and machine learning techniques to analyze the relationship between demographic factors and insurance charges.

B. Project Goal

The objective of this project is to investigate factors that influence health insurance charges using demographic and lifestyle data from the Kaggle dataset [3]. The project evaluates the performance of Linear Regression and Random Forest models, identifies key predictors through statistical tests and permutation importance, and provides insights into the factors influencing premiums.

II. METHOD

A. Statistical Testing and Model Formulation

To enhance interpretability and identify statistically significant predictors, we performed p-value testing using the Ordinary Least Squares (OLS) method. Predictors with p-values below 0.05 were considered statistically significant contributors to the target variable (`charges`). This approach complements machine learning methods, ensuring transparency in model behavior.

Two models and one LLM were implemented:

- 1) **Linear Regression:** Provides a baseline for comparison and ensures interpretability.
- 2) **Random Forest Regressor:** Captures non-linear relationships and ranks feature importance.
- 3) **Preliminary Data Analysis:** Ola El-Shiekh's Kaggle notebook provided inspiration and methods for performing exploratory data analysis, including visualizations and feature examination, which were adapted for this study. [4].
- 4) **ChatGPT Assistance:** ChatGPT was utilized to generate initial code suggestions, refine analysis and Random Forest approaches, explore statistical modeling ideas, and revise grammar issues. [5].

The models were evaluated using Mean Squared Error (MSE) and R^2 scores.

III. RESULTS

A. Correlation Analysis

The correlation heatmap (Fig. 1) illustrates the relationships among numerical features. Smoking status showed the strongest positive correlation with insurance charges, followed by BMI and age. These findings align with our p-value testing results, confirming the critical role of these factors in determining insurance charges.

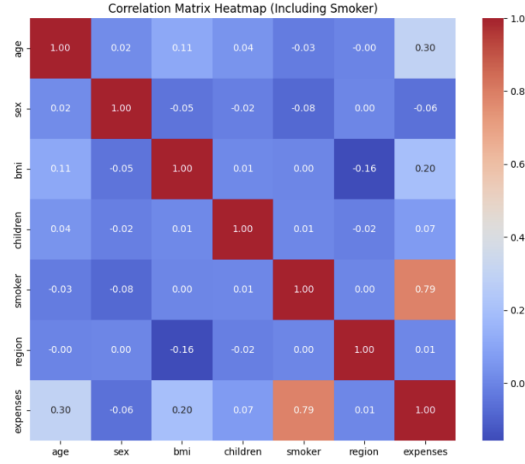


Fig. 1. Correlation heatmap of numerical features, including smoking status.

B. Permutation Importance (Random Forest)

The permutation importance plot (Fig. 2) ranks the predictors in the Random Forest model based on their contributions to reducing the model's R^2 . Smoking status (`smoker`) is the most important predictor, followed by BMI (`bmi`) and age (`age`). These results align closely with the findings from OLS regression.

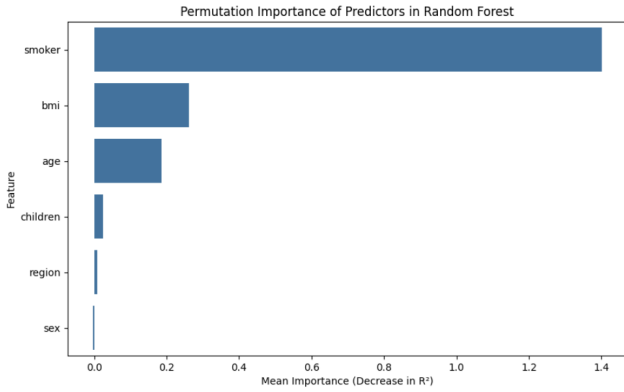


Fig. 2. Permutation importance of predictors in the Random Forest model.

C. P-Value Rankings (OLS Regression)

Table I ranks the predictors from OLS regression by their p-values (sorted in ascending order). Predictors with smaller p-values have stronger statistical significance. The `smoker` variable is highly significant ($p < 10^{-16}$), followed by `age` and `bmi`, confirming their substantial influence on health insurance charges.

TABLE I
OLS PREDICTORS RANKED BY P-VALUE (ASCENDING ORDER)

Predictor	P-Value	Significance
smoker	$< 10^{-16}$	Highly Significant
age	2.89×10^{-89}	Highly Significant
bmi	1.42×10^{-31}	Highly Significant
children	5.10×10^{-4}	Significant
region	2.01×10^{-2}	Significant
sex	6.94×10^{-1}	Not Significant

D. Model Performance

The performance of both models is summarized in Table II. Random Forest significantly outperformed Linear Regression, achieving higher accuracy and lower error.

TABLE II
MODEL PERFORMANCE

Model	MSE	R^2 Score
Linear Regression	33,639,075.09	0.78
Random Forest Regressor	21,395,177.85	0.86

IV. DISCUSSION

The results from both methods provide valuable insights into the predictors of insurance expenses:

- **Agreement Between Methods:** Both Random Forest and OLS highlight `smoker`, `bmi`, and `age` as the most influential predictors, indicating their critical role in determining health insurance charges.
- **Statistical Significance:** P-value testing from OLS identifies `smoker` as the most statistically significant predictor ($p < 10^{-16}$), aligning with its top rank in permutation importance.
- **Non-Linear Relationships:** Random Forest captures non-linear interactions, particularly for `bmi` and `age`, enhancing predictive accuracy compared to linear methods.

The combined use of statistical testing and permutation importance offers a robust framework for analyzing predictors, balancing interpretability and model complexity.

V. CONCLUSION

This study demonstrates the complementary roles of statistical and machine learning methods in predicting health insurance charges. Key findings include:

- Smoking status, BMI, and age are consistently identified as the most influential predictors by both methods.
- P-value testing enhances interpretability, while permutation importance captures complex interactions.
- Random Forest achieves higher predictive accuracy, but OLS provides valuable statistical insights.

Future research could explore fairness metrics to ensure equitable pricing and extend this framework to incorporate additional health-related features.

REFERENCES

- [1] S. of Actuaries, "Social and other determinants of life insurance demand," <https://www.soa.org/resources/research-reports/2022/determinants-life-insurance/>, 2022, accessed: 2024-12-02.
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- [3] A. Kaggle, "Insurance charges dataset," <https://www.kaggle.com/datasets/awaiskaggle/insurance-csv>, 2024, accessed: 2024-12-02.
- [4] O. El-Shiekh, "Insurance eda," <https://www.kaggle.com/code/olaelsheikh/insurance-eda>, 2024, accessed: Dec. 2024.
- [5] ChatGPT, "Code suggestions for importing packages and random forest modeling," <https://openai.com/chatgpt>, accessed: December 5, 2024.

APPENDIX

Figures

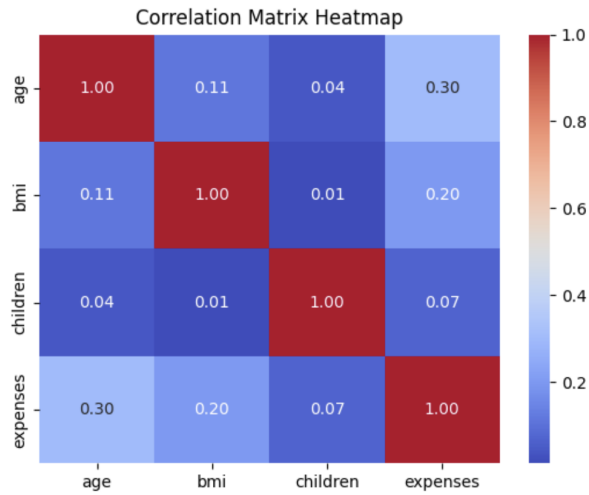


Fig. 3. Correlation Matrix Heatmap.

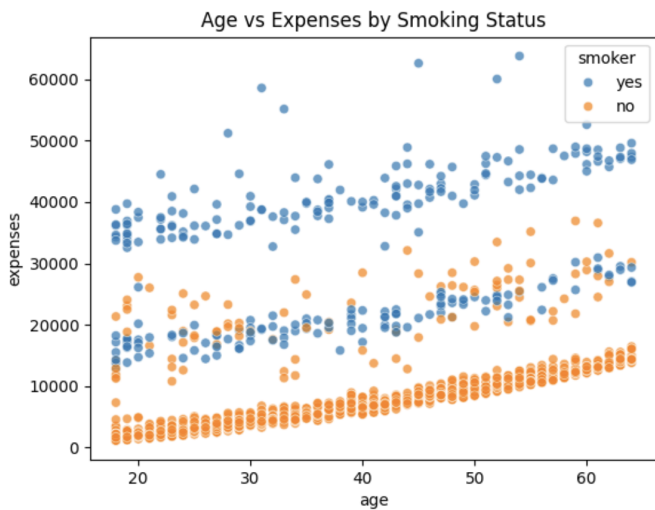


Fig. 4. Age vs Expenses by Smoking Status.

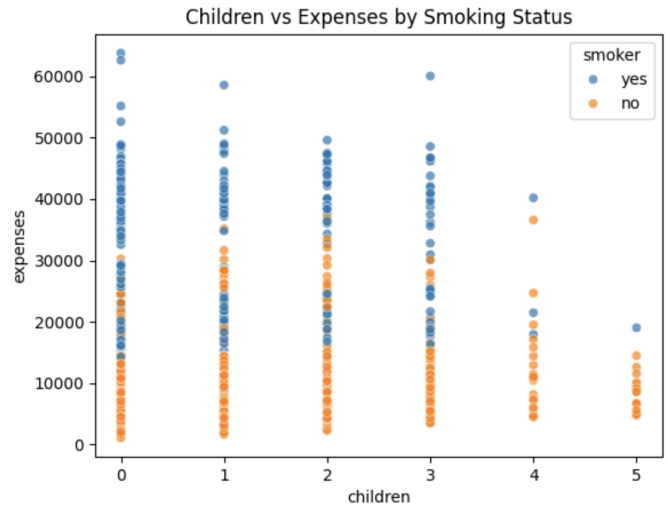


Fig. 5. Children vs Expenses by Smoking Status.

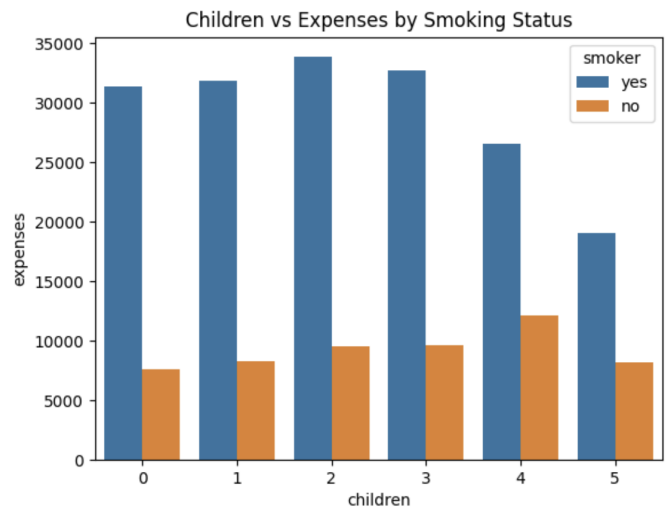


Fig. 6. Distribution of Smoking Status by Number of Children.

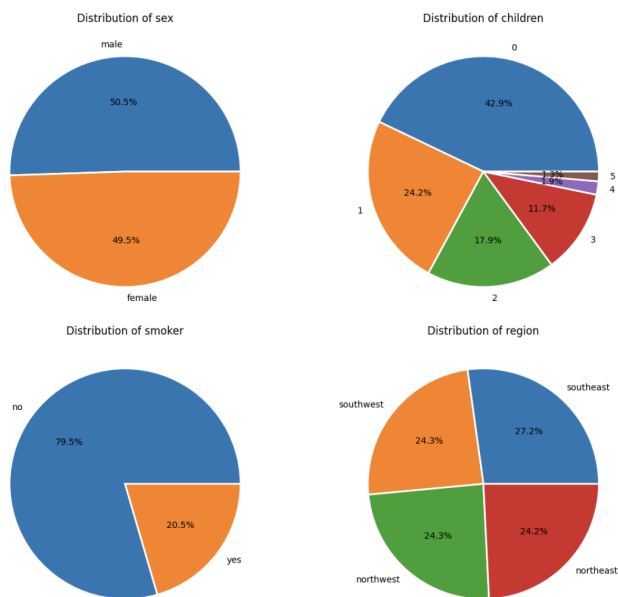


Fig. 7. Age Distribution for Smokers vs Non-Smokers.

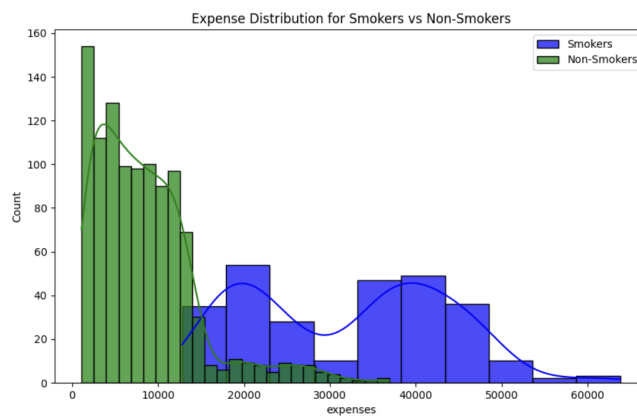


Fig. 9. Children vs Smoking Status Crosstab.

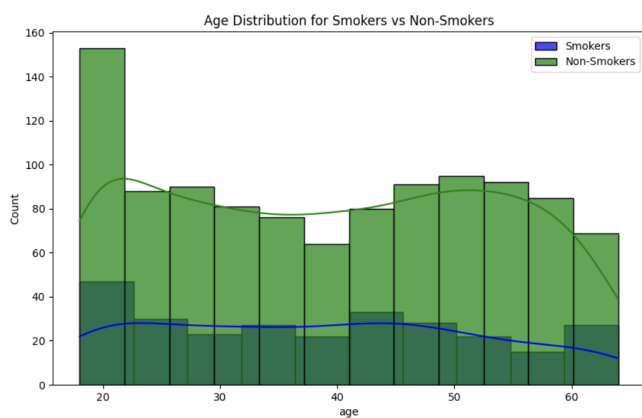


Fig. 8. Expense Distribution for Smokers vs Non-Smokers.

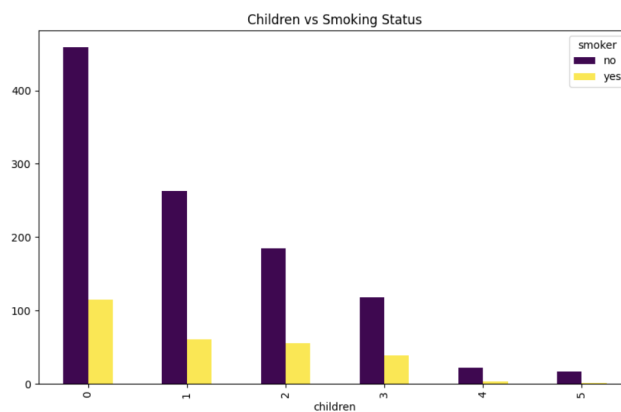


Fig. 10. Comparison of Age, BMI, and Expenses Between Smokers and Non-Smokers.

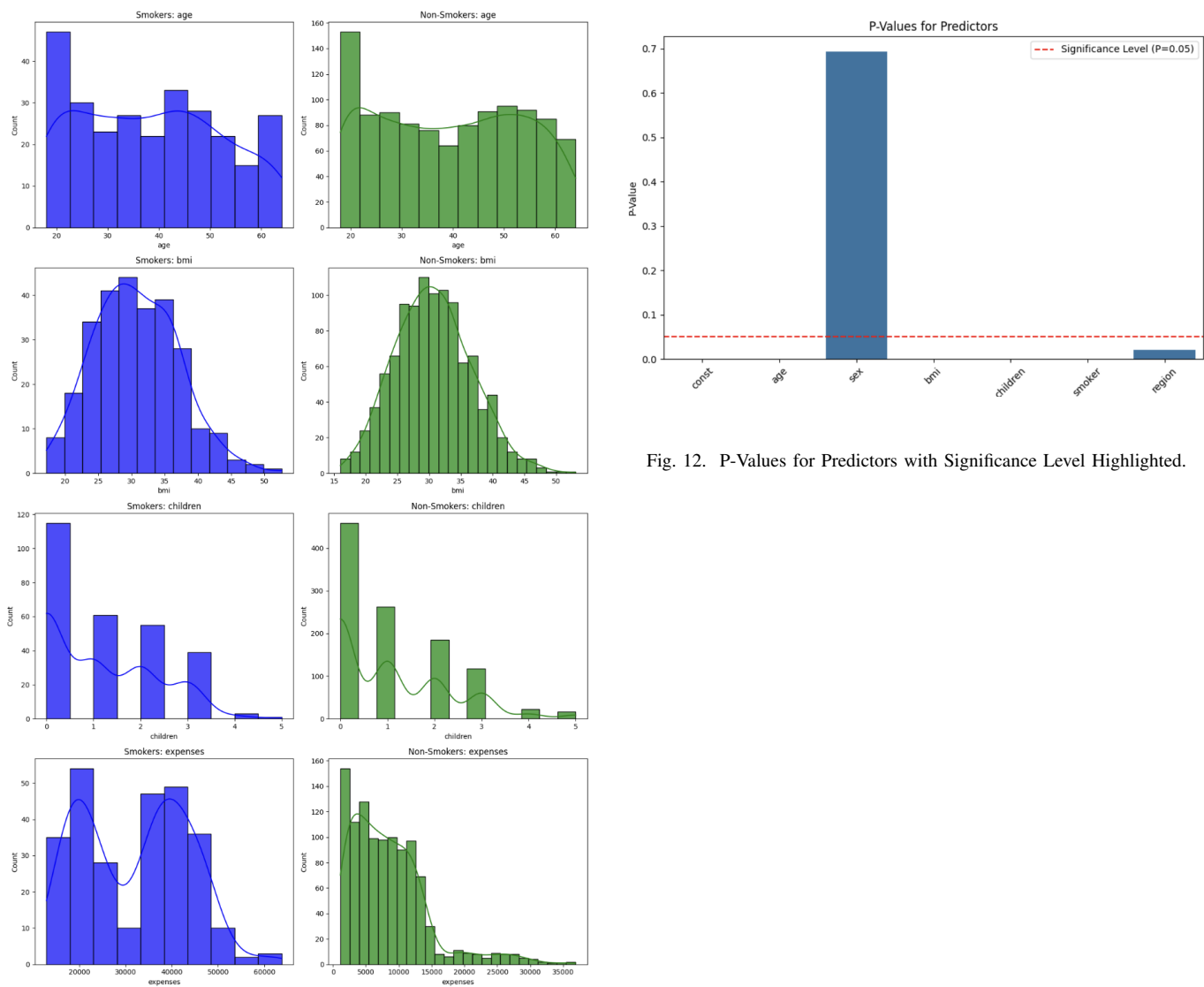


Fig. 12. P-Values for Predictors with Significance Level Highlighted.

Fig. 11. Combined Histogram of Residuals for the Model.